Ingyu Yoo

Ph.D. Candidate

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SUMMARY

Ph.D. candidate in Materials Science and Engineering with expertise in Transmission Electron Microscopy (TEM). Specializes in the structural characterization of 2D materials, focusing on multilayer systems and heterostructures. Employs data science for data-driven research and applies hypothesis-driven methodologies based on materials science principles to elucidate material properties and behavior at the nanoscale.

EDUCATION

Convergence Major — Seoul National University, South Korea Sep. 2024 – Aug. 2026
Graduate School of Data Science · Co-Advisor: Prof. Taesup Kim

Integrated M.S. & Ph.D. — Seoul National University, South Korea Mar. 2022 – Present
Department of Materials Science and Engineering · Advisor: Prof. Miyoung Kim

B.S. — Yonsei University, South Korea Mar. 2016 – Feb. 2022
Department of Materials Science and Engineering

WORKING EXPERIENCE

Visiting Researcher — Monash University, Australia Dec. 2025 – Feb. 2026
Department of Materials Science and Engineering · Advisor: Prof. Jian-min Zuo

Visiting Researcher — University of Illinois Urbana-Champaign, USA Feb. 2025 – Dec. 2025
Department of Materials Science and Engineering · Advisor: Prof. Jian-min Zuo, Prof. Pinshane Huang

Undergraduate Intern — Seoul National University, South Korea Jan. 2021 – Feb. 2022
Department of Materials Science and Engineering · Advisor: Prof. Miyoung Kim

Undergraduate Intern — Yonsei University, South Korea Mar. 2020 – May 2020
Department of Materials Science and Engineering · Advisor: Prof. Heon-Jin Choi

RESEARCH INTERESTS

Stacking-Sequence Determination in Multilayer TMDs using 4D-STEM
Developed a methodology that resolves stacking sequences layer-by-layer from scanning nanobeam diffraction, and correlated the resolved sequences with electronic properties measured by Kelvin probe force microscopy (KPFM).

Displacement Vector Field Mapping in Epitaxially Grown TMD Heterostructures
Characterized structural variations, including Moiré lattice defects, in CVD-grown MoS_2/WSe_2 heterostructures, and linked strain relaxation to the evolution of these defects through displacement vector field mapping with 4D-STEM.

Distinct Atomic Reconstruction Pathways across the Fabrication-Architecture Landscape
Showed that releasing out-of-plane freedom in capping-free WSe_2 promotes curved atomic reconstruction pathways, revealing how reduced vertical confinement reshapes relaxation in moiré bilayers.

Identified attractive and repulsive soliton-network interactions that emerge when an artificial stack is assembled on a CVD-grown single domain.

Machine Learning and Deep Learning for TEM Analysis
Implemented a saliency-based clustering model that extracts the coordinates of every atomic column in atomic-resolution STEM images, and used those coordinates for quantitative strain and defect analysis.

Developing a unified analysis platform spanning (S)TEM imaging, EDS, EELS, and 4D-STEM, integrated with LLM agents to enable task-aware research automation.

SKILLS

(S)TEM Experiment and Analysis

- Imaging (TEM/STEM/4D-STEM) and Spectroscopy (EDS/EELS)
- Currently focusing on comprehensive analysis via STEM (HAADF) and 4D-STEM Dataset
- Advanced License: (Thermo Fisher Scientific) Tecnai F20, Talos, Themis Z / (JEOL) JEM-2100F

Machine Learning and Deep Learning based methodologies for TEM analysis

- Using Convolutional Neural Network for defect identification (Classification of Dopants and Vacancies)
- Atom center fitting in atomic resolution STEM image via Attention model and Clustering
- Various ways for Denoising in low-dose imaging — Clustering, Autoencoder, Filtering optimization via Genetic Algorithm, etc.

Data Analysis / Programming Skills

- Proficient in Python, C, and C++, with working knowledge of Big Data management system queries
- Conducted relational data analysis with Neo4j and managed large-scale data using PostgreSQL queries
- Built a multi-agent system to forecast the electric-vehicle battery market from hypothetical news scenarios
- Applied deep reinforcement learning (DQN) to optimize trades in prediction markets

PUBLICATION

1. Jinwoo Kim‡, **Ingyu Yoo‡**, Jaewoong Joo‡, Hyoungseok Lee, Jundong Yeo, Jian-Min Zuo, Miyoung Kim*, Gwan-Hyoung Lee*, “Visualizing Lattice-mismatch-dependent Strain Relaxation in Epitaxially Grown MoS2/WS2 and MoS2/WSe2 Hetero-bilayers”, Rep. Prog. Phys., (2026)
2. **Ingyu Yoo‡**, Jinwoo Kim‡, Gwan-Hyoung Lee, Miyoung Kim*, “Unravelling the Diverse Stacking Sequence in Polar Layered Materials using Diffraction Polarity”, ACS Applied Materials & Interfaces, (2025)

ONGOING PROJECTS (lead author, to be submitted)

- **I. Yoo** et al., Moiré Lattice Defects in van der Waals heterostructures of MoS2/WSe2
- Y.Chang, **I.Yoo**, J.Ryu et al., Curved Atomic Reconstruction Pathways under Reduced Vertical Confinement in Moire Bilayers
- **I. Yoo**, H. Kim et al., Inequivalence of solitons and effects on reconstructed networks in artificially stacked 3R trilayer MoS2
- C. Shin, **I.Yoo** et al., Quantitative Atomic Structure Analysis Using Saliency-Based Clustering in STEM
- Y. Tak, **I. Yoo**, J. Seo et al., TARAE: Task-Aware Research and Analysis Environment for Electron Microscopy – An Open-Source, Cross-Platform Agentic Characterization Suite
- **I. Yoo**, J. Huh et al., Probing Dopant Distribution and Chemical State in Potassium-Doped Tungsten

Full publication list: https://scholar.google.com/citations?user=Ylbr8n8AAAAJ&hl=en

CONFERENCE

- (Poster) Ryu, J., S. Lee, S. Kim, **I. Yoo**, Y.-C. Joo, M. Kim*, “A machine-learning approach to characterization of amorphous materials with EELS-SI and 4D-STEM”, 2022 Materials Research Society Spring Meeting, Hawaii, USA (2022)
- (Oral) Yunyeong Chang, **Ingyu Yoo**, Ayoung Yuk, Hyobin Yoo, Miyoung Kim*, “Understanding Twisted Transition Metal Dichalcogenide Domain Node Atomic Transition”, 2022 Materials Research Society Spring Meeting, Hawaii, USA (2022)
- (Oral) **Ingyu Yoo**, Jinwoo Kim, Jusang Lee, Gwan-Hyoung Lee, Miyoung Kim*, “Unraveling stacking seqeunce of WSe2 spiral structure using nanobeam diffraction”, The 20th International Microscopy Congress, Busan, South Korea (2023)
- (Oral) **Ingyu Yoo**, Miyoung Kim*, “Diverse Stacking Sequences Discovery in Transition Metal Dichalcogenides (TMDs) through Diffraction Polarity Analysis”, The 2024 Annual Spring Conference of Korean Society of Microscopy, Hongcheon, South Korea (2024)
- (Poster) Yoonsang Tak, Sehwan Song, Hyeonyu Kim, **Ingyu Yoo**, Sang Woon Hwang, Sungkyun Park, Dooyong Lee, Sangmoon Yoon, Miyoung Kim*, “Structural origins of magnetic property evolution in epitaxial ilmenite–hematite solid solutions”, The 2024 Annual Spring Conference of Korean Society of Microscopy, Gyeongju, South Korea (2024)
- (Oral) **Ingyu Yoo**, Jinwoo Kim, Gwan-Hyoung Lee, Miyoung Kim*, “Determination of Stacking Sequence and its correlation to properties on Multi-Layer Transition Metal Dichalcogenides”, 2024 Materials Research Society Fall Meeting, Boston, USA (2024)
- (Oral) **Ingyu Yoo**, Jinwoo Kim, Jaewoong Joo, Gwan-Hyoung Lee, Miyoung Kim*, “Structural Responses to Interlayer Coupling in Aligned Heterostructure”, 13th Asia-Pacific Microscopy Congress, Brisbane, Australia (2025)
- (Oral) **Ingyu Yoo**, Yoonsang Tak, Jongho Lee, Mireu Kim, Miyoung Kim*, “Quantitative Atomic Structure Analysis Using Saliency-Based Clustering in STEM”, 13th Asia-Pacific Microscopy Congress, Brisbane, Australia (2025)

- (Poster) Jinwoo Kim, **Ingyu Yoo**, Jaewoong Joo, Miyoung Kim, Gwan-Hyoung Lee*, "Complex Strain Distribution in Van der Waals Epitaxy of Transition Metal Dichalcogenides Heterostructures with Lattice Mismatch", 23rd International Nano Technology Exhibition, KINTEX, South Korea (2025)
- (Oral) **Ingyu Yoo**, Jinwoo Kim, Jaewoong Joo, Gwan-Hyoung Lee, Miyoung Kim*, "Deciphering Moiré Periodicity Modification in van der Waals Heterostructure", Microscopy and Microanalysis 2025, Salt Lake City, USA (2025)
- (Poster) Eunseo Oh, **Ingyu Yoo**, Chanyoung Shin, Ahjeong Lyu, Kyuree Kim, Miyoung Kim*, "Machine Learning-Based Quantitative Microstructure Analysis for Integrity Evaluation of IN718 Alloys", The 2026 Annual Spring Conference of Korean Society of Microscopy, Daejeon, South Korea (2026)
- (Oral) **Ingyu Yoo**, Hangyel Kim, Pinshane Y. Huang, Arend M. van der Zande*, Miyoung Kim*, "Atomic Reconstruction in a Single-Twist Trilayer MoS2 on a CVD-Grown Bilayer", The 21th International Microscopy Congress, Liverpool, United Kingdom (2026)
- (Invited Talk) **Ingyu Yoo**, Yunyeong Chang, Jinseok Ryu, Jinwoo Kim, Seong Chul Hong, Ji-Hwan Baek, Jaewoong Joo, Jian-min Zuo, Hyobin Yoo, Gwan-Hyoung Lee, Miyoung Kim*, "Interlayer lattice relaxation and moiré structural evolution in TMDs", The 21th International Microscopy Congress, Liverpool, United Kingdom (2026)

UPCOMING CONFERENCE

- (Poster) Jinseok Ryu*, **Ingyu Yoo**, Yoonsang Tak, Miyoung Kim, "TARAE: Task-Aware Research and Analysis Environment for Electron Microscopy – An Open-Source, Cross-Platform Agentic Characterization Suite", 2026 Materials Research Society Fall Meeting, Boston, USA (2026)
- (Oral) Hangyel Kim, **Ingyu Yoo**, Pinshane Y. Huang, Arend M. van der Zande*, Miyoung Kim*, "Distinct Domain Boundaries Reveal Polarization-Switching Pathways in Trilayer Sliding-Ferroelectric MoS_2", 2026 Materials Research Society Fall Meeting, Boston, USA (2026)

AWARD / FUNDING / SCHOLARSHIP

- Best Presentation Award, Symposium "Carbon-Based Materials/2D Materials", 20th International Microscopy Congress, Busan, South Korea (2023)
- Outstanding Oral Presentation Award, The 2024 Annual Spring Conference of Korean Society of Microscopy, Hongcheon, South Korea (2024)
- BK21 FOUR International Joint Training Program for Outstanding Graduate Students, Ministry of Education & National Research Foundation of Korea (NRF) (2025–2026)
- Full Tuition Scholarship for 6 consecutive semesters, Seoul National University (2022–2024)
  - 2022 Spring, 2022 Fall: Teaching Assistant Scholarship
  - 2023 Spring, 2023 Fall: BK21 Scholarship
  - 2024 Spring: Teaching Assistant Scholarship
  - 2024 Fall: Academic Excellence Scholarship

TEACHING EXPERIENCE

Workshop Lecturer — Korean Society of Microscopy Jul. 25, 2024

29th TEM Workshop, Jul. 23–26, 2024 · Lecture: Practical Training in Data Processing Techniques

Developed and delivered a Python programming lecture tailored for Transmission Electron Microscopy (TEM) data analysis, focusing on foundational Python skills and practical applications. Provided step-by-step examples for effective TEM data handling, analysis, and visualization.

Invited Seminar — Monash University Feb. 03, 2026

Monash Centre for Electron Microscopy Talk · Title: Decoding Structural Complexity in van der Waals layered structures

This talk presents two types of layered systems grown by chemical vapor deposition (CVD): (i) WSe2 multilayers exhibiting diverse stacking sequences, and (ii) MoS2/WSe2 heterobilayers containing Moiré lattice defects.

Invited Seminar — Seoul National University May 2026

AI in Physics seminar, May 14 & 21, 2026, Department of Physics, SNU · Lecture 1: Deep learning in TEM / Lecture 2: Application of TEM analysis to 2D materials

Introduction to deep learning in transmission electron microscopy, atom center fitting using pyramid saliency model (showcase in 2D materials), stacking sequence determination in multilayer TMDs using diffraction polarity, displacement vector field mapping in moiré lattice defects, resolving atomic arrangement using multi-slice ptychography.

Teaching Assistant — Seoul National University Mar. 2022 – Aug. 2026

- Physical Chemistry of Materials 1 — Spring 2022, 2023, 2024. Q&A via email and office hours, review sessions, exam and assignment support.
- Electric, Optical, and Magnetic Properties of Solids — Fall 2022, 2023. Q&A via email and office hours, review sessions, exam and assignment support.
- Comprehensive Materials Laboratory — Spring 2023, Fall 2023, Spring 2024, 2026. Led selection of graduation research topics, explained experimental data and theoretical background, and gave detailed feedback on papers and posters.

  Research Topics: 'Enhancing the Clarity of Atomic-Resolution TEM Images for Few-Layer WSe2 Systems through Genetic Algorithm-Based Denoising' (Spring 2023, Fall 2023); 'Correlative study on rhombohedral stacked TMDs' stacking sequence and its properties' (Spring 2024); 'Atomic origins of Moiré lattice defects in MoS2-WSe2 hetero-bilayers' (Spring 2026).